

Sensory Notebook

Living with sensory sensitivity — research, and what has actually worked

2026-08-12

A public research notebook and personal record. Research summaries cite their sources; personal experience is labelled as experience and is never presented as a finding. Nothing here is medical advice.

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About this document

This is a consolidated export of the Sensory Notebook, a public repository that does two things at once and tries never to blur them: it records longform, sourced answers to research questions about sensory processing, and it records one person's own sensory profile and the interventions he has actually tried.

Keeping those apart is the point. Every entry below carries a banner saying which kind it is. Research summaries cite sources; personal experience does not need a citation and is never presented as a finding.

This is not medical advice. It is written by someone who is not a clinician. Describing what an intervention does is not the same as telling you to use it.

Sensory terminology is contested. Where a term belongs to a particular framework — DSM-5-TR, ICD-11, the occupational-therapy literature — the entries name the framework rather than using the terms interchangeably.

Generated 2026-08-12 from the repository as it stood on that date.

Part I — The profile

Personal experience. Something actually tried, in the real world, with a verdict. Not evidence, not a finding, and not a recommendation to anyone else.

Sensory profile

Relevant context

I have ADHD. Everything below is described against that background rather than as a separate condition, and where the distinction matters the notebook tries to say which is which rather than assume.

Per modality

For each: what the trigger is, how it presents, how severe, what makes it worse or better.

Auditory

Primary trigger: intelligible background speech, while I am trying to concentrate.

- Background conversation is the problem — people talking nearby, someone on a phone call in the same room. Overheard *speech* specifically, not noise in general and not loudness.
- It is **state-dependent, and that is the defining feature**. If I am relaxing in the same room, with the same people having the same conversations at the same volume, it does not bother me at all. It only becomes a problem when I am trying to focus on something.
- I cannot tune it out. It is not that I notice it and then let it go; the filtering step other people appear to have does not happen for me.
- Open-plan offices are inherently very challenging for this reason.
- **Other people's television, while I am studying or working nearby.** The longest-running instance of this — it was a real problem growing up and it is still a problem today. TV audio is speech, and it arrives while I am trying to focus, so it fits the pattern above. But I escalate to earplugs specifically when it is *loud enough*, which the pattern above says should not be the deciding variable.
- My hearing itself is normal. I have had hearing testing done and it came back completely normal — see *story.md*.

Severity rating still to fill.

What I actually do about it, in escalation order. This ladder is the practical core of the notebook, and each rung has its own entry:

1. **Close the door** — *recommendations/closing-doors.md*. Free, first resort, works where I control the door.
2. **Earplugs** — *recommendations/custom-moulded-earplugs.md*. The general go-to. Custom moulds in preference to *Loop* or anything off the shelf, on comfort.
3. **Earmuffs over the earplugs** — *recommendations/earmuffs.md*. Tried, not adopted, less comfortable.

Where this sits diagnostically, and the candidate mechanisms: *answers/260812-background-speech-and-focus.md*.

Second trigger: intrusive noise at home. This is not the same problem as the one above, and the two should not be run together.

- **It is not state-dependent and it is not about concentration.** The speech problem only exists when I am trying to focus. This one does not switch off when I stop working.

- **The defining feeling is loss of control over my environment.** Home is supposed to be the safe space from the noise of the city, and when the noise gets in, what is gone is not quiet — it is the sense that the space is mine. That framing matters more than the decibels.
- **Recorded instances**, both from Jerusalem apartments — see *story.md*:
 - **Honking at night**, in the previous apartment, audible in the bedroom and while working. Extremely stressful; got to the point where my daily happiness was at stake.
 - **Drilling and loud construction work**, in the current apartment, ongoing.
- **Separately: a screaming baby.** Since becoming a father in July 2025 this has been really challenging. Recorded here because it is real, but it is not obviously the same phenomenon as the city-noise instances and is not being filed as one.

Severity rating, and the relationship between this trigger and the speech trigger, still to establish. Note that neither has been described in terms of volume as such — one is about intelligibility, the other about intrusion — so “loudness sensitivity” is not currently a supported description of either.

What I use for this: *recommendations/custom-moulded-earplugs.md*, and *recommendations/loop-earplugs.md*.

Visual

To fill.

Tactile

To fill.

Olfactory / gustatory

To fill.

Vestibular / proprioceptive

To fill.

Interoceptive

To fill.

Interactions

Which sensitivities compound which others, and under what conditions (fatigue, hunger, stress, medication timing, time of day).

To fill. Specifically unrecorded so far: whether the auditory effect above changes with stimulant medication, time of day, or tiredness.

Environments

Where it’s worst and where it’s manageable — home, office, street, transit, shops, family gatherings.

- **Open-plan office — worst.** Inherently very challenging. Named as the clearest case.
- **Any shared space while relaxing — fine.** Explicitly not a problem; see the state-dependence note under Auditory.

Street, transit, shops and family gatherings: to fill.

The story

What it felt like before there was a name for it

The thing I actually noticed was narrow and specific: when I am trying to focus on something, background conversation destroys it. Not noise — conversation. People talking near me, someone taking a phone call in the same room. Open-plan offices are inherently very challenging for me, and for years that was just a fact about me with no explanation attached.

The earliest version of it I can point to is other people's television while I was trying to study. That was a real problem growing up, and it is not a childhood memory — it still happens, and I still deal with it the same way, by closing doors and then reaching for earplugs if it is loud enough. Whatever this is, it has been the same shape for a very long time.

What made it hard to place is that it is not a general intolerance of sound. In the same room, with the same people, at the same volume, if I am relaxing rather than working, it does not bother me at all. That asymmetry is the whole shape of the thing, and it is the part that made it feel like it could not be a "hearing" problem.

What it got misattributed to

Hearing, first. I have had hearing testing done and my hearing came back completely normal. Which closed one door without opening another — a normal test result told me what it was not, and nothing about what it was.

Beyond that, I do not know whether it is a specific pathology of its own or simply part of the overall ADHD picture. That is the open question this notebook started with.

What changed, when

Ten years in Jerusalem, across several apartments — and every one of them has been noisy in a different way. There is a pattern to how it happens: you view a place, it is pretty quiet, you move in, and you find out you viewed it at a quiet time of day. That has happened to us recurrently, which is its own small lesson about how little a viewing tells you.

Undated, during the previous apartment — honking at night. It was audible in the bedroom and audible while I was working. What made it different from the background-speech problem was that it had nothing to do with concentration: it was the feeling of not having control of my environment, of the place that is meant to be a safe space from the noise of the city being invaded by the noise of the city. Not insanity, but extremely stressful, and it got me looking for literally any solution under the sun.

Around 2023, in response to the above — the audiology appointment where I had custom earplugs made, after trying Sony over-ear ANC headphones, ANC earbuds and Loops.

The part of this worth recording is not the product. It is that **I had struggled with this for many years and it was never a priority**. It never felt worth a hundred dollars. What changed was not new information — it was that my daily happiness was at stake. I suspect that arithmetic is what keeps a lot of people from buying the thing that would help them, and the hundred dollars turns out to be the cheap part; the years of deferring it were the expensive part.

I had also gone in for a product and come out with something else as well. The reaction I got was that this is something they see all the time and custom moulds are a perfect solution for it — and being somewhere where the thing was completely ordinary was, in itself, incredibly validating. That is one of the pivotal points in the whole story. Years of working around something quietly, and then a room where it is a routine appointment.

Something similar happened from the other direction with Loop earplugs. As they became popular, friends I would never have associated with sensory issues started telling me they had found them life-changing. The plugs themselves I ended up not wearing much once I had the customs — but the discovery that these people were out there, and had been all along, was its own thing.

The earplugs themselves, and the question they left open, are in *recommendations/custom-moulded-earplugs.md* and *recommendations/loop-earplugs.md*.

Recently — the current apartment, where there is drilling and very loud construction work going on. We have not finished unboxing. I am very glad the earplugs are in the cabinet.

August 2026 — first attempt at actually looking it up rather than living with it. See *questions/260812-background-speech-and-focus.md* and the answer alongside it.

The comparison that made it click

I asked other people whether it bothers them, including my wife. What came back was not “yes, a bit less” — it was a different relationship to the sound entirely. She described being able to deliberately tune *in* and hear somebody speaking, as a thing she chooses to do. Whereas I cannot tune it *out*.

That is a subtle difference and I think a vastly telling one. Same room, same sound, same physical input — and the difference is not sensitivity, it is which direction the control runs. For her the default is filtered and attending is the act. For me the default is unfiltered and there is no act available.

If that generalises at all, it is a useful thing to be able to say out loud to neurotypical people, because “background noise bothers me” gets heard as “I am fussy about volume,” which is not the claim.

What I wish someone had told me earlier

To fill.

Part II — Questions and answers

Research summary. Drawn from the literature. Every non-obvious claim is traceable to a file in the source list at the end of this document.

Why does background conversation only wreck my focus when I'm trying to focus?

Asked 2026-08-12 · answered 2026-08-12 · confidence: medium · tags: auditory, adhd, attention, workplace

The question

Why does background conversation only wreck my focus when I'm trying to focus?

I have ADHD, and I've always had a specific kind of sensitivity to sound that can be roughly described like this: when I'm working, trying to focus on something, I find that background sound — background conversations in particular — really puts my focus across. Open-office designs are inherently very challenging for me for that reason.

I don't know if this is a specific pathology, or a recognised part of the overall ADHD picture.

I've had hearing testing done and organically my hearing is totally normal.

And strangely, it's very focused, very specific for me — by which I mean, if I'm relaxing in a space I can hear people having phone calls and conversations and it doesn't bother me at all. It's only when I'm trying to focus that it's hard.

I've asked other people, including my wife, about whether this bothers her at all. She can actually tune *in* and hear somebody speak, whereas I can't tune it *out*. I think that's a subtle but vastly telling difference in how maybe neurotypical and neurodiverse people can experience the exact same sensory input from a totally different perspective.

So: how might this be defined? Is it pretty typical of ADHD, and what specific mechanism drives it?

The answer

Background speech and focus: what it's called, whether it's an ADHD thing, and what drives it

Not medical advice. This is a literature summary written for a personal notebook, by someone who is not a clinician. See the disclaimer in the *README*.

Short answer

1. **It has a name, and the name is not a diagnosis.** The effect where task-irrelevant background speech degrades performance is called the **irrelevant speech effect**, or more accurately the **irrelevant sound effect (ISE)**. It is a robust laboratory phenomenon found in everyone, not a pathology.
2. **Yes, it is bigger in ADHD, and that is measured, not folklore.** Adults with ADHD take a larger hit from irrelevant sound than matched controls, and the size of the hit tracks self-rated inattention.
3. **The state-dependence is not strange — it is predicted.** The ISE attacks verbal serial short-term memory specifically. When you are not running a verbal-serial task, there is nothing for it to corrupt. Sitting in a room relaxing is not a verbal-serial task. Writing, reading, coding and holding a mental thread are.

4. **The trigger is speech structure, not loudness.** In the office-acoustics literature the performance decrement scales with **speech intelligibility (STI)**, not sound level. This is why noise can help while conversation destroys you, and it is not a contradiction.
 5. **Your normal audiogram is exactly what all of this predicts.** Nothing in this account involves your ears.
 6. **What is genuinely unsettled is the mechanism** — specifically whether the deficit is an early sensory-filtering failure or a late attentional-control failure. The evidence points both ways and the field has not resolved it.
-

1. The phenomenon has a name

The **irrelevant sound effect** is the reliable impairment of immediate serial recall by task-irrelevant sound that participants are explicitly instructed to ignore. It has three properties that matter for your question.

It is speech-preferring. In the largest recent test — *Leist, Lachmann & Klatte (2025)*, 137 children and 98 adults — narrative speech impaired verbal recall more than environmental sounds (dog barking, bells, siren) did, and environmental sounds still beat silence. Speech is the worst case, not merely one case.

It is not about volume. That study ran at ≈ 59 dB LAeq — ordinary conversational level — and still produced a large effect ($\eta^2p = .197$).

It is graded by intelligibility. The office-acoustics literature has quantified this. *Haapakangas, Hongisto & Liebl (2020)* systematically reviewed 11 experiments manipulating the **Speech Transmission Index** and confirmed a dose–response curve: decline begins above **STI ≈ 0.21** , and the decrement **exceeds 15% above STI 0.6**. Where speech is audible at all, its effect relates more to intelligibility than to sound level.

That last point is the single most useful fact in this answer. The damaging quantity is *how comprehensible the speech is*, not *how loud it is*.

2. Is it a recognised part of the ADHD picture?

Yes, in two separate senses that are worth keeping apart, because they come from different literatures and make different claims.

Sense one: distractibility is literally an ADHD diagnostic criterion

DSM-5-TR inattention criterion **A1h** reads: “*Is often easily distracted by extraneous stimuli (for older adolescents and adults, may include unrelated thoughts).*” So the symptom you are describing is not an add-on to ADHD, it is one of the nine inattention items. The parenthetical was added for DSM-5 and makes explicit that the distracting stimulus can be internal as well as external.

Note what the criterion is a claim about: **attention**, not perception. The DSM does not credit ADHD with any sensory criterion. Autism does have one — hyper- or hyporeactivity to sensory input, criterion B4 — and ADHD does not. See *spd-diagnostic-status.md*.

Sense two: the effect has been measured, and it is larger in ADHD

Pelletier et al. (2016) is the directly on-point study: 32 adults with ADHD and 32 controls, serial recall in quiet vs with irrelevant sound they were told to ignore.

- The ADHD group recalled worse in **both** conditions — some of the gap is baseline short-term memory, not distraction.
- The **extra** decrement from irrelevant sound was **larger** in the ADHD group.
- Performance under irrelevant sound **correlated with self-reported attention problems**.
- Sustained attention looked intact — the sound penalty *shrank over the course of the task*.

Alongside that, a separate literature finds elevated sensory hypersensitivity in adult ADHD generally, with the **auditory modality most affected** (*Bijlenga et al. 2017*), and elevated self-reported sensory-gating problems that correlate with the ASRS inattention factor (*Micoulaud-Franchi et al. 2015*). In that last study the ADHD group scored higher on the Sensory Gating Inventory than not only healthy controls but also

patients with schizophrenia, and the authors' phrasing is almost a restatement of your complaint: attention "involuntarily drowned by many irrelevant environmental stimuli."

So: typical of ADHD? Yes. Both as a diagnostic criterion and as a measured group difference. What none of that establishes is that it is *only* ADHD, or that ADHD explains its particular shape in your case.

3. What it is *not* — and why ruling these out is informative

Getting the label wrong here is not pedantry; each wrong label points at a different literature and a different set of interventions.

Not hyperacusis

Hyperacusis is discomfort or **pain** at sound levels most people tolerate, driven by **loudness**, measurable as reduced loudness discomfort levels. It is present whether or not you are doing anything. Yours is level-independent and vanishes when you stop working. Wrong construct. (ICD-11 does list hyperacusis, under AB7Y — see *decreased-sound-tolerance-terminology.md*.)

Not misophonia

Misophonia is an intense **emotional** reaction — anger, disgust — to **specific trigger sounds**, characteristically body sounds, at any volume. It is stimulus-bound and emotionally loaded. Your description contains no anger and no specific trigger, and the trigger set is "any intelligible conversation," which is a category, not a trigger. Misophonia is also **not in ICD-11 or DSM-5-TR** at all.

Not straightforwardly APD

"Normal audiogram plus trouble with speech and background sound" is the classic route to an **auditory processing disorder** referral, and there are two reasons to be careful. The first is that APD's validity as a distinct entity is heavily contested — no consensus definition, no gold-standard test, and critics argue most APD test batteries are effectively tests of language and attention. It overlaps so heavily with ADHD that the standard tests cannot separate them, and the field is drifting toward the softer term "listening difficulties." See *apd-diagnostic-validity.md*.

The second reason is sharper and specific to you. **APD is difficulty extracting the signal you want out of noise. Yours is difficulty discarding a signal you do not want while doing something else.** Those are close to opposite failures. You have not reported struggling to follow a conversation in a noisy bar. You have reported being unable to *stop* following one. An APD battery is not built to detect that.

Not "sensory processing disorder"

SPD is not in DSM-5-TR and not in ICD; the American Academy of Pediatrics recommended against using it as an independent diagnosis in 2012. The SOR literature can usefully **describe** a sensory difference; it does not provide a thing to *have*.

4. Candidate mechanisms

Five accounts are live. They are not mutually exclusive, and the honest position is that the field has not decided between them. What follows is what each predicts, so you can see which ones your experience actually discriminates.

(a) Interference-by-process — the phonological/serial-order account

Speech is **changing-state**: it carries automatic, obligatory order information. If your current task also requires holding items in serial order — reading a sentence, composing one, tracking a line of argument — the two order representations collide. This is **pre-attentive and obligatory**. You cannot decide not to do it.

Evidence: Leist et al. found adult **spatial** serial recall completely immune while verbal recall was disrupted. Domain-general accounts predict both should suffer. The authors read the adult immunity as favouring **phonological-store interference**.

Prediction: effort does not help. Only removing or degrading the speech helps.

(b) Attentional capture

Background speech intermittently grabs attention away from the task. Distinct from (a) and — per the **duplex-mechanism account** (Hughes et al. 2013) — **controllable by top-down task engagement**, whereas interference-by-process is not.

Evidence: in Leist et al., children's spatial recall was disrupted, by speech and non-speech equally, and the disruption disappeared by fourth grade. Read as attentional capture waning as attentional control matures.

Prediction: effort and engagement do help, at least partly.

(c) Sensory gating failure

A pre-attentive filter that normally attenuates repeated irrelevant input. Indexed by **P50 suppression** in a paired-click paradigm. Micoulaud-Franchi et al. report reduced P50 suppression in adult ADHD, correlating with self-reported gating problems.

Prediction: the failure is early, sensory, and upstream of anything voluntary.

(d) Load theory — a late, cognitive-control failure

Lavie's account: perceptual capacity is limited and allocated involuntarily. High perceptual load exhausts it on the task and starves distractors; low load leaves spare capacity that spills onto whatever else is around.

Forster et al. (2014) found adults with ADHD more distracted than controls — **but** that raising perceptual load reduced distraction **equally well in both groups**. If the ADHD deficit were in early perceptual selection, load should have helped the ADHD group differently. That it did not points the deficit at **late selection / cognitive control**.

Note the direct conflict with (c). The gating literature says the failure is early and sensory; the load-theory result says it is late and control-based. Both are in print, and this notebook is not in a position to adjudicate. Flagging it rather than picking a side.

(e) Moderate Brain Arousal / stochastic resonance — why noise can help

Söderlund, Sikström & Smart (2007): white noise **improved** cognitive performance in children with ADHD and **impaired** it in controls — a cross-over interaction. The MBA model attributes this to stochastic resonance, with the optimum noise level modulated by dopamine, so lower-dopamine individuals need more external noise to reach it.

This resolves the apparent paradox of “noise helps me, people talking wrecks me.” White noise is stationary and unstructured: no order cues, no phonology, nothing for mechanisms (a) or (b) to bite on. It just raises the arousal floor. Speech is non-stationary and structured, and the structure is what does the damage.

Caveat, and it is a real one: replications are mixed, and the stochastic-resonance mechanism was directly challenged in a 2024 *Neuropsychologia* paper. The **contrast** between structured speech and unstructured broadband noise is well supported by several independent literatures; the **dopaminergic explanation** for the benefit is contested. Lean on the first, not the second.

5. Why only when you're trying to focus — the crux

This is the part of your question the literature answers best, and the answer is that your observation is not strange at all. It falls out of the findings in two layers.

Layer one — the trivial but real part. “Distraction” is only defined relative to a task. With no task there is no performance to degrade, so there is nothing to notice. This is true but unsatisfying, because it would apply to anyone, and your point is that it does *not* apply to everyone equally.

Layer two — the substantive part. The ISE is not a general “sound bothers me” effect. It targets **verbal serial short-term memory**. Leist et al. is the clean demonstration: adults doing a *spatial* serial task at 59 dB

of Danish newsreader showed **no disruption at all**, while adults doing a *verbal* task at the identical sound level were reliably impaired.

Map that onto your two states:

State	Verbal-serial load	Predicted ISE
Working — reading, writing, composing, holding a thread	High	Large
Relaxing in the same room	~none	~none

Same room, same people, same volume, opposite outcome — **because the variable that changed is not the sound, it is whether you are currently running the one cognitive process that overheard speech corrupts**. Your “strangely, it’s very specific” is the literature’s central finding, not an anomaly in it.

There is a third layer worth flagging, and it is where I am least confident. *Sörqvist & Marsh (2015)* argue that **concentration shields against distraction** — the more engaged you are, the less background processing happens. Read naively that predicts the opposite of your experience. The reconciliation I would offer is that the shield is a property of the **engaged state**, not of the **attempt to become engaged**: the vulnerable window is exactly the effortful not-yet-locked-on period, and if ADHD means spending more time in that window and less time inside the shield, the phenomenology follows. (**speculation**) — this reconciliation is mine, not a claim any of these papers makes.

6. “She can tune in; I can’t tune out”

Your framing is sharper than the literature’s usual phrasing, and mostly it holds up.

What is supported. In typical listeners, selective attention *enhances* cortical tracking of the attended talker and *suppresses* tracking of the ignored one — the auditory cortex ends up representing the attended speech and not the ignored speech. The default state is filtered, and attending is an operation performed on top of it. That is exactly the asymmetry you described from the other side.

What is inference, not evidence. The natural extension — that in ADHD the *separation* between attended and ignored streams is reduced, so the ignored stream stays represented — is a plausible prediction and it is consistent with the gating findings. But I looked and **did not find ADHD-specific neural speech-tracking studies** to support it. The behavioural claim (ADHD shows a larger ISE, weaker self-reported gating, reduced P50 suppression) is supported. The neural-tracking bridge is **my extrapolation**. Treat it as a hypothesis, and as a good candidate for a follow-up question.

One caution on the comparison itself. A single person’s introspective report about their own attention is not data, in either direction — introspective access to one’s own filtering is poor, and “I can tune in when I want” may describe the experience of a shield working rather than a mechanism. The comparison is a good *illustration*; it is not evidence. It is recorded as personal narrative in *context/story.md*, not as a finding.

7. What follows from this

Framed as implications of the findings, not as recommendations — those belong in *recommendations/* once something has actually been tried.

- **Optimise intelligibility down, not volume down.** The STI dose–response means the target variable is comprehensibility. Absorption, screens, distance from talkers and masking sound all lower STI. Masking *raises* total sound level while *lowering* STI, which is why it works and why “just make it quieter” is optimising the wrong quantity.
- **Structured vs unstructured is the distinction that matters** when choosing what to put in your ears. The evidence separates broadband/stationary sound from speech-structured sound; it does not license any particular product claim.
- **Masking has a ceiling.** Spatial release from masking means individual voices can stay intelligible up to roughly four talkers, and in office simulations low masking levels made multi-talker environments *worse*.

- **Enclosure changes what coping is even possible.** In a 1,078-worker survey, open-plan occupants resorted to headphones; shared-office occupants could take breaks, change workspace, close a door. Annoyance was more likely above 20 occupants and without acoustic treatment. This is the empirical shape of “open-plan is inherently challenging.”
- **Individual susceptibility is documented** (Ellermeier & Zimmer, 1997), which is the citable basis for treating this as an individual accommodation question rather than a general office-design preference.
- **If the dominant mechanism is (a), effort is not a lever.** Interference-by-process is not controllable by trying harder. If it is (b), engagement helps. Worth attending to which one your own experience looks like, since they imply completely different strategies.

8. Confidence and what's still open

High confidence: the ISE exists, prefers speech, is graded by intelligibility not loudness, and targets verbal serial memory. The state-dependence you described follows directly from that.

Medium confidence: the effect is larger in ADHD. One good direct study (Pelletier, n=64) plus converging evidence from adjacent measures. Not a large literature.

Low confidence: which mechanism dominates. The gating evidence (early/sensory) and the load-theory evidence (late/control) genuinely conflict. Also low: the MBA/dopamine explanation for why unstructured noise helps.

Not established at all, flagged as speculation: the engagement-window reconciliation in §5, and the neural-tracking account in §6.

Caveat on sourcing across the board: with the exception of Leist et al. 2025 (open access, read in full record), everything here was read at abstract or secondary-summary level. Each file in *sources/* carries its own provenance note saying which.

Follow-up questions worth asking

- ☐ Does neural speech-tracking research show reduced attended/ignored separation in ADHD? (Directly tests §6.)
- ☐ Does stimulant medication change ISE magnitude? Secondary sources claim methylphenidate normalises some sensory hypersensitivities — unverified. This is also open question 3 in *context/scope.md*.
- ☐ Does the habituation Pelletier found (sound penalty shrinking over the task) match lived experience, or is that a lab artefact of short sessions?
- ☐ What is the actual measured STI of a typical open-plan desk, and what would it take to get one below 0.21?
- ☐ Is there evidence separating interference-by-process from attentional capture in ADHD specifically? That is the question that would decide whether effort is a lever.

Is the right earplug attenuation for a screaming baby the same as for sleep?

Asked 2026-08-12 · answered 2026-08-12 · confidence: medium · tags: auditory, hearing-protection, products, parenting, safety, israel

The question

Is the right earplug attenuation for a screaming baby the same as for sleep?

I have two pairs of custom moulded earplugs from an audiologist — a soft silicone pair for sleeping and a hard acrylic pair for daytime. Background in *recommendations/custom-moulded-earplugs.md*.

I became a father in July 2025, and the screaming-baby end of it was really challenging. I went back to the audiologist looking for a pair that would take the edge off without being unsafe — meaning, without cutting me off from the baby. What I was told was that the silicone sleep pair I already have is what I should use,

and they would not sell me another pair even though I would have bought one. I thought that was a good answer and I have no reason to doubt it.

But I do not want to assume it is correct just because it was generous. So:

1. **Is the attenuation you want for sleep actually the same as the attenuation you want for parenting?** These feel like different requirements to me. For sleep the goal is to hear as little as possible. For parenting the goal is to still hear the baby clearly — just not at full intensity. Those could easily want different amounts, or different *shapes* of attenuation across frequency, rather than the same plug in both cases.
2. **What is actually unsafe here, and what is just uncomfortable?** I framed the request as “take the edge off without being unsafe”, but I do not know where the real line is — for the baby’s sake or for my hearing’s.
3. **The red-alert siren constraint.** Living in Israel, this is the one that actually worries me. During wartime the last place you want to be is somewhere you cannot hear an emergency alert. What level of attenuation still leaves a red-alert siren reliably audible, including asleep? This is a geographically specific problem and I am aware most readers will not have it, but for anyone here it decides the whole question.

Status

Answered 2026-08-12 — *answers/260812-earplugs-for-parenting-vs-sleep.md*.

Points 1 and 2 are answered: the advice holds, and sleep is the harder case of the two, so reusing the sleep pair errs in the safe direction. **Point 3 is not answered numerically** — the Israeli siren specification is not publicly published, and the answer says so instead of substituting a plausible figure.

The answer

Parenting versus sleep: the advice holds, but not for the reason you were given

Not medical advice, and not safety certification. This is a literature summary written for a personal notebook by someone who is not a clinician or an acoustician. See the disclaimer in the *README*.

Short answer

1. **The audiologist was right, and the reasoning is better than “reuse what you have”.** For a normal-hearing listener, a hearing protector attenuates the signal *and* the background by roughly the same amount, so the **signal-to-noise ratio at the eardrum is preserved**. Earplugs do not selectively delete the baby. This is measured, not theoretical: across 80 listeners and seven warning signals, wearing hearing protection **improved** audibility for normal-hearing listeners.
2. **The two cases do differ — and sleep is the harder one.** Detecting a sound while awake and being woken by it are different thresholds, and the sleeping threshold is higher and individually unpredictable. So a plug you have validated against *sleep* is **conservative in the right direction** when used awake. The advice you were given errs safe.
3. **“Unsafe” cuts both ways, and the direction you weren’t worried about is the real one. Overprotection** is a recognised occupational failure mode: too much attenuation degrades communication, causes missed signals, and — the self-defeating outcome — makes people take the protection off. The target is a **band, not a maximum**.
4. **The crying is loud enough to be worth protecting against.** Measured levels from crying children fall between **99 and 120 dB(A)**, and the authors of that study independently recommend earplugs for parents.
5. **The siren limb cannot be answered numerically.** The Israeli siren specification is not publicly published, in English or Hebrew. Anyone who gives you a dB figure for it is probably quoting a burglar alarm.

Why preserved signal-to-noise is the whole answer

The intuition behind the original worry — that blocking sound might cut you off from the baby — imports a model where an earplug is a switch. It is not. It is a subtraction applied to everything arriving at the ear.

If the cry is 100 dB and the room is 40 dB, a 20 dB plug gives you an 80 dB cry in a 20 dB room. The **60 dB gap between them is unchanged**. What decides whether you detect the cry is not the absolute level but whether it clears your **masked threshold** — audibility against the background — and that relationship survives attenuation nearly intact for a normal-hearing ear. *sources/hpd-warning-signal-audibility.md*

This is why the railway study found HPDs *improving* audibility for normal-hearing listeners rather than degrading it. It also explains where the argument fails: for **hearing-impaired** listeners it reverses, because the attenuated signal can drop below an already-elevated absolute threshold. The authors propose a usable criterion — security is assured where average absolute threshold at 500/1000/2000 Hz in the better ear stays **below 30 dB HL**. Your hearing tested completely normal (*context/profile.md*), which places you in the group the finding applies to, not the exception.

Why sleep is the harder case, not the easier one

This is the part that actually answers “are these different requirements”.

Being woken is governed by the **auditory arousal threshold**, which is not the same quantity as your waking detection threshold — and there is **no correlation between the two**. A normal audiogram tells you nothing about how easily you wake. Roughly 55–70 dBA at the pillow wakes most adults, regulations generally specify **75 dBA at the pillow** for bedroom alarms, and even so **20% of adults in one study slept through a ten-minute alarm**. *sources/sleep-arousal-thresholds-and-alarms.md*

So the ordering is: **arousal threshold > waking detection threshold**. A plug specified so that a baby can still wake you is, by construction, a plug through which you can hear a baby while conscious. Going the other way — validating for parenting and assuming it covers sleep — would be the unsafe direction, and is not what you were advised to do.

(Speculation, flagged: nobody has measured arousal thresholds through earplugs. Subtracting an attenuation figure from an arousal threshold is a reasonable inference and not a finding. It is the weakest structural step in this answer.)

The direction of “unsafe” you were not asking about

You framed the request as *take the edge off without being unsafe*, meaning: don’t block so much that I miss something. The occupational literature has a name for exactly that concern — **overprotection** — and treats it as a real failure mode rather than a theoretical one: missed safety signals, degraded communication, feeling “out of touch”, slower responses, and protectors being removed in order to hear. *sources/hpd-overprotection-and-derating.md*

Two consequences:

- **In moderate noise, 10–15 dB of real attenuation is typically all that is needed**, and guidance frames the goal as a protected level in the **70–80 dBA** region — a floor as well as a ceiling.
- **Labelled attenuation is not obtained attenuation**. OSHA halves the NRR; NIOSH subtracts 25% for muffs, 50% for foam plugs and **70% for all other earplugs** — the category custom moulds fall into. Whatever number is on the box, assume you are getting substantially less, which incidentally makes over-blocking less likely than the box suggests.

So the audiologist declining to sell you a heavier pair is consistent with the hygiene literature, not just commercially decent.

The siren: what can and cannot be said

Cannot: there is no public specification for the sound-pressure level of the Home Front Command siren network. Searched in English and Hebrew on 2026-08-12 across press, Hebrew Wikipedia and civil-defence

material, and it is not there. The 115 dB and 125 dB figures that surface are **commercial burglar-alarm sounders** — a different product class entirely. *sources/israeli-siren-level-not-published.md*

Can, structurally: this exact problem is what **ISO 7731** exists for. Its introduction states that correctly designed danger signals reliably call attention to a hazard **even when hearing protection is worn**, and its normative Annex B gives the method for calculating the **effective masked threshold** — a quantity whose definition explicitly folds in hearing-protector attenuation as a known input. The standard’s position is that a protected listener is a normal design case, not an excluded one. *sources/iso-7731-auditory-danger-signals.md*

Can, factually about the system: the Israeli alert is **multi-channel by design** — public sirens, the Home Front Command app, the national emergency portal, radio and TV interruption, and location-based SMS with loud audio notification, staged from 15–30 minutes ahead down to roughly 90 seconds before impact. The siren is one path among several, and its own documentation concedes coverage is incomplete in some areas. That is a description of how the system is built, not a recommendation about what to rely on.

The honest bottom line on this limb: the general acoustic argument says a normal-hearing person wearing moderate attenuation is not thereby deaf to a loud low-frequency external siren. But that is reasoning from principle with **the key input missing**, and it is about a *sleeping* listener, where the evidence is weakest. This notebook cannot responsibly turn that into an assurance, and does not.

What would change this answer

- **Knowing which plugs you actually own.** A *solid* custom plug is roughly 25–30 dB; a *filtered* one is typically 9, 15 or 25 dB. Nothing in the notebook records whether yours are solid or filtered, and that single fact changes every number here. *sources/custom-earplug-attenuation-and-filters.md*
- **The siren specification**, if it can be obtained from the Standards Institution of Israel, the civil-defence regulations, or the 104 hotline.
- **Any study of arousal thresholds through hearing protection.** None was found. If one exists it replaces the weakest step in this answer.

One thing worth knowing about the materials

Silicone versus acrylic is a **comfort, seal and durability** distinction. The **filter, not the shell material, sets the attenuation curve** — no source found gave a direct acoustic comparison between the two materials. So “the soft pair” and “the hard pair” are not inherently different amounts of protection, which makes the advice to reuse the sleep pair less surprising than it first sounded.

What is the ideal level of attenuation for studying or working — and does it change if you are more sensory-overloaded than I am?

Asked 2026-08-12 · answered 2026-08-12 · confidence: low · tags: auditory, hearing-protection, products, attention, workplace

The question

What is the ideal level of attenuation for studying or working — and does it change if you are more sensory-overloaded than I am?

The companion question to *260812-earplugs-for-parenting-vs-sleep.md*. That one is about safety limits — how much you can block and still hear a baby or a siren. This one is about the opposite end: what you should actually be aiming for when the goal is concentration.

1. **Is there an ideal attenuation level for focused work, and is it a maximum?** My instinct is to assume more blocking is better, but I do not know that. The profile’s primary trigger is intelligible background speech (*../context/profile.md*), and if the problem is intelligibility rather than volume, then the target might be “drop speech below the point where it resolves into words” rather than “make it as quiet as possible”. Those are different specs and they might want different products.

2. **Does the answer differ by severity?** I am interested in what this looks like for someone *more* sensory-overloaded than I am, not just for me. Whether the right level scales with how badly affected you are, or whether there is a single acoustic target everyone is aiming at, seems like a real question rather than an obvious one.
3. **Is there a point where blocking too much becomes its own problem?** Silence, occlusion effect, hearing your own pulse and chewing — anecdotally these bother some people. Where that sits relative to the useful range is not something I know.

Why this is not obvious from the products

Every earplug is sold on a single NRR or SNR number, which is one figure for a thing that varies by frequency. The *existing recommendations* are all decided on comfort, because comfort is what determined whether the plug got worn — none of them involved comparing attenuation curves. So the notebook currently has strong opinions about wearability and no data at all about levels.

Status

Answered 2026-08-12 — *answers/260812-ideal-attenuation-for-focused-work.md*, with confidence: low, because the main finding is that the number does not exist.

Limb 1 has no published figure and the question turns out to be aimed slightly wrong — the predictive variable is intelligibility, not level, and that is a room property rather than an earplug property. Limb 2 is **unstudied**, not disputed. Limb 3 has three real effects but no threshold.

The answer

Ideal attenuation for focused work: the number does not exist, and the question is aimed slightly wrong

Not medical advice. A literature summary for a personal notebook, written by someone who is not a clinician or an acoustician. See the disclaimer in the *README*.

Short answer

1. **There is no published ideal attenuation figure for concentration.** Not “it is disputed” — it has not been established. No study found measured attenuation against cognitive performance. The confident numbers online come from earplug manufacturers.
2. **More is not better, and that part is well supported.** Overprotection is a recognised failure mode with a stated target *band* rather than a maximum. Hearing protection has an optimum.
3. **The variable that predicts performance loss is intelligibility, not level** — which means an earplug is a slightly indirect instrument for the problem in your profile.
4. **Whether it scales with severity is unstudied.** Every study found used unselected adults. The question is not contested; it is unasked.
5. **Yes, there is a cost to over-blocking** — increased listening effort, and one small study found worse mood and higher anxiety while occluded. But there is also an **acclimatisation effect**, so first impressions understate a plug.
6. **The flatness of the curve probably matters more than the number on the box.**

Why the number does not exist

The honest state of the evidence, and the reason this answer carries confidence: low despite being fairly definite:

- Research on hearing-protection devices found **all tested HPDs degraded performance on most tasks** versus the open ear, specifically by **increasing listening effort** — cognitive resources spent on hearing

are not available for the task. That is measured in conditions where you still need to monitor sound; in silent solo work it should largely not arise, but that boundary is reasoned, not tested.

- One pilot study — **thirty adults, two hours** — found visual attention **improved** during occlusion and stayed elevated afterwards, while **state anxiety rose and mood fell**. The same review notes prior studies were **inconsistent**, some finding nothing.
- Almost everything else asserting that earplugs aid concentration is **manufacturer content** — Loop, CURVD, Snugs, Hearprotek. It leans on the solid *noise impairs cognition* literature and then makes an untested leap to *therefore earplugs improve productivity*.

sources/hpd-cognitive-performance.md

This notebook carries a Loop recommendation. That recommendation stands on its own terms — it records what happened when you used them. It is not evidence, and the company’s marketing is not evidence either, and keeping those apart is the point of the repository.

The question is aimed slightly wrong, and that is the useful finding

The whole quantitative apparatus for background speech and work performance is built on **STI — the Speech Transmission Index**, not on decibels. What predicts performance loss is **how intelligible** the speech is, not how loud. Performance decline begins around **STI $\approx$ 0.21**, and where speech is audible at all its effect relates more to intelligibility than to level. *sources/hongisto-haapakangas-sti-performance.md*

ISO 3382-3 fixes two thresholds: **distraction distance** at STI 0.50 and **privacy distance** at STI 0.20, where speech falls under 30% word intelligibility. The graded experiments cluster low — a cut-off near 0.23, performance unaffected between 0.00 and 0.30, effort required to resist distraction from about 0.26 upward. So the target is **around 0.2, not 0.5**. *sources/iso-3382-3-distraction-and-privacy-distance.md*

The catch: STI is a property of a room and a listening position, not of an earplug. There is no defined way to convert an attenuation figure into an STI change, so “aim for STI 0.2, therefore buy an N dB plug” is a step the literature does not license.

This is why your own profile note reads the way it does. If intelligibility rather than volume is the operative variable, then a plug — which lowers level roughly uniformly and leaves the speech/background ratio intact — is attacking the problem side-on. The levers that actually move STI are **absorption, screens, distance, and masking sound**, and masking *raises* total level while *lowering* intelligibility. Your instinct to close a door (*recommendations/closing-doors.md*) is, in these terms, a better-targeted intervention than the earplugs are.

(Speculation, flagged: this reasoning also predicts that an earplug should help less with the speech-while-concentrating trigger than with the honking-at-night one — which is exactly the distinction your profile draws for independent reasons. Suggestive, not demonstrated, and the notebook does not currently record whether the plugs help with the speech case at all.)

Is more blocking better? No, and here is the shape of the ceiling

Overprotection is a documented occupational failure mode: degraded communication, missed signals, feeling “out of touch”, slower response, and protectors removed so people can hear. In moderate noise **10–15 dB of real attenuation is typically sufficient**, with target protected levels quoted in the **70–80 dBA** region — a floor as well as a ceiling. *sources/hpd-overprotection-and-derating.md*

Against that, **labelled attenuation is not obtained attenuation**. NIOSH derates non-foam earplugs by 70%. So the practical risk of accidentally over-blocking with an off-the-shelf number is lower than the box implies — and the practical value of a *fitted* plug is that its delivered attenuation is more predictable, not that it is higher.

Does the answer differ if you are more affected?

Unstudied. Every study located used unselected adults; none stratified by sensory over-responsivity, and none looked at ADHD populations.

What is documented is that **individual susceptibility to the irrelevant speech effect varies**, and that variation is measured, not anecdotal — already catalogued in the existing STI source. That is the empirical

hook for saying a population-average acoustic target under-protects some people. It supports “*the right target is not the same for everyone*”. It does not tell you which way, or by how much, and it is about room acoustics rather than hearing protection.

So the second limb of the question is a genuine open research question, and worth stating as one rather than answering thinly.

Is there a point where blocking too much is its own problem?

Three real effects, none quantified against a threshold:

- **Listening effort**, above — a cost paid whenever you still need to hear something.
- **Mood and anxiety**, from the thirty-person pilot. One small study; recorded because it is the only signal found in either direction, not because it is solid.
- **Flatness of the curve**. The steeper the attenuation slope across frequency, the more the protected ear differs from the open ear. Germany’s IFA codes protectors for this: **code W** requires a mean slope no greater than **3.6 dB/octave**, **code X** an extremely flat ≤ 2 **dB/octave**. Foam is the bad case — it kills highs far harder than lows, which is why everything sounds muffled through it. *sources/hpd-warning-signal-audibility.md*

Practical reading: choose flat over strong. A 15 dB flat filter and a 25 dB steep plug are not two points on one scale; the first leaves the world sounding like itself, quieter. Filters around **15 dB are reported to be the flattest**, and a solid unfiltered custom plug is **25–30 dB** — the blunt end of the range. *sources/custom-earplug-attenuation-and-filters.md*

The finding that vindicates how this notebook already chose

Every recommendation in this repository was decided on **comfort**, with no attenuation figure compared — noted in *recommendations/README.md* before any of this research was done. Two results say that was not a lazy proxy:

- **Overprotection’s characteristic outcome is people removing the protection.** Wearability is not a soft criterion; it is the one the safety literature keeps arriving at.
- **Acclimatisation is documented:** pianists’ reported degradation dropped between a first and second performance. So judging a plug on first wear systematically underrates it — which is worth knowing before rejecting one, and is the closest thing to actionable advice this answer contains.

What would close the gap

- **Which filter, if any, is in the custom plugs.** Still unrecorded, still the single most useful missing fact.
- **Any study measuring attenuation against cognitive performance.** None found; if it exists, it answers the question as posed.
- **Anything stratifying acoustic targets by sensory over-responsivity.** Would answer limb two, which is currently unanswerable rather than merely unanswered.

Part III — What has actually worked

Personal experience. Something actually tried, in the real world, with a verdict. Not evidence, not a finding, and not a recommendation to anyone else.

Closing doors

kind: technique · verdict: works · since: unknown · modality: auditory

Closing doors

Personal experience, not evidence. Not medical advice — see the disclaimer in the *README*.

The problem

Someone else's television, audible from where I am trying to study or work. This started in childhood and it has not gone away — it is still my situation today when it happens.

The specific case is a household one: other people are doing something entirely reasonable in their own space, and it lands in mine.

What was tried

Close the door.

That is the whole intervention, and it is worth writing down precisely because it is the one nobody writes down. It is free, it takes two seconds, it requires no purchase and no appointment, and it is the first thing I reach for before anything else on this list.

What happened

It works, often enough to be the default. Where it works it is better than every other option here, because there is nothing in your ears.

What it did not fix

- **It fails on volume.** If it is loud enough, a closed door is not enough, and I escalate to earplugs — *custom-moulded-earplugs.md*.
- **It only works where you control the door.** It is useless in an open-plan office, which is the single worst environment on record in *../context/profile.md*, and useless anywhere the sound and you are in the same room.
- **It is a social act, not just an acoustic one.** You are closing a door on people. In a shared home that carries a meaning beyond the decibels, and the cost of that is real even when it is small. This is the part that makes it harder than it sounds.

Transferability

- **Try this before buying anything.** The cheapest intervention in this notebook, and the one most likely to be skipped over on the way to a product.
- **It generalises to any door you control** — but note that the reason it works is probably not only attenuation. A door also removes the sightline and the sense of the activity happening at you. Which of those is doing the work here is not established.
- **Treat it as step one of a ladder, not a solution on its own.** The full escalation in practice is: close the door → earplugs → (earmuffs over the top, tried and not adopted, see *earmuffs.md*).

Custom moulded earplugs, made by an audiologist

kind: product · verdict: works · since: 2023-00 · cost: ~USD 100 for both pairs · modality: auditory

Custom moulded earplugs, made by an audiologist

Personal experience, not evidence. Nothing on this page is a research claim, and the attenuation questions it raises are deliberately left open — see [questions/260812-earplugs-for-parenting-vs-sleep.md](#).

Not medical advice. See the disclaimer in the *README*.

The problem

We have lived in Jerusalem for ten years, and **every apartment has been noisy in a different way**. There is a pattern to how this happens: you view a place, it is pretty quiet, you move in — and it turns out you viewed it at a quiet time of day. That has happened to us recurrently.

In the previous apartment what emerged after moving in was **honking, at night**. We could hear it clearly in the bedroom, and I could hear it while working.

The thing that made this different from the background-speech problem is that **it was not about concentration**. It was the feeling of not having control of your environment. The place that is supposed to be your safe space from the noise of the city is suddenly invaded by the noise of the city. I would not go so far as to say it drove me to insanity, but it was extremely stressful, and it got me to the point of looking for literally any solution under the sun.

Why it took so long to buy anything. This is something I had struggled with for many years, and it was never a priority. It never felt worth a hundred dollars. What changed was not new information — it was that my daily happiness was at stake. That is worth naming, because I suspect the same arithmetic keeps a lot of people from buying the thing that would help them.

What was tried

In roughly this order, during the same desperate search:

1. **Over-ear ANC headphones — Sony**, model not recorded and not considered important.
2. **In-ear ANC earbuds.**
3. **Loop earplugs.** Worked, but superseded on comfort — separate entry: [loop-earplugs.md](#).
4. **Custom moulded plugs from an audiologist**, taken from impressions of my own ears. Two pairs, deliberately in different materials:
 - **Silicone — soft.** For sleeping.
 - **Acrylic — hard.** For daytime use.

The two-material split was the point. A soft plug is what you can lie on; a hard plug is what you want when you are upright and moving around.

What happened

They work, and they are one of the most useful things I have ever bought. These are the ones I wear all the time, in preference to everything else on that list.

The comfort is the whole thing, and it is not a minor point. Earplugs can become their own sensory problem — generic ones feel dry and uncomfortable sitting in your ear, so the cure introduces a new irritation. The customs are **by far the most inert** of anything I have tried. I am aware they are in my ears, but they do not nag at me. (I keep them washed, as you would with any earplug, but that is hygiene, not comfort.)

The appointment itself was a pivotal moment, separately from the product. Being somewhere where the request was completely ordinary — the audiologists' reaction was along the lines of *this is something we see all the time, it's a perfect solution* — was **incredibly validating** in itself. That is not a feature of the earplugs. It is a feature of walking into a place where the thing you have been quietly working around for years is a routine Tuesday.

The parenting follow-up. I became a father in July 2025. A screaming baby was really challenging, and I went back looking for a pair that would take the edge off without being unsafe. The audiologist's answer was

that **the silicone sleep pair I already owned was what I should use** — and declined to sell me another pair even though I would have bought one. I appreciated that a lot.

They kept earning it after the original problem went away. In the current apartment there is drilling and very loud construction work going on. We have not finished unboxing and I am already very glad these are in the cabinet. A thing bought for one apartment's noise has turned out to be a thing you own for the next one's.

What it did not fix

- **It did not settle whether sleep-grade attenuation is the right attenuation for parenting.** The advice I was given was to reuse the sleep pair, and I have no reason to doubt it — but “the pair you already own” and “the pair with the right curve for this situation” are two different claims, and only the first one was actually tested. Filed as an open question rather than assumed: *questions/260812-earplugs-for-parenting-vs-sleep.md*.
- **It did not solve the red-alert siren problem.** Living in Israel, there is a hard constraint on any hearing protection worn at home: during wartime, the last place you want to be is somewhere you cannot hear an emergency alert. Custom plugs do not come with an answer to that, and I do not have one. It is a geographically specific problem, but for anyone here it is the binding one.
- **They are a mitigation, not control of your environment.** They made the honking survivable. They did not give me back the thing that had actually been taken, which was a bedroom that was quiet because it was quiet.
- **Whether they help with the primary auditory trigger is not recorded.** The profile's main problem is intelligible background speech while concentrating, which is a filtering problem rather than a loudness problem. Whether these plugs do anything for it, in an open-plan office or otherwise, is simply not something this file knows.

Transferability

For someone else to get the same result:

- **You need an audiologist who takes ear impressions and makes custom moulds.** This is not a mail-order product; the fit is the entire value.
- **The gain is comfort, not attenuation.** If off-the-shelf plugs already sit comfortably in your ears and you wear them, custom moulds buy you much less. The before/after here is “unwearable” to “wearable, all day, by choice” — not “quiet” to “quieter”. If you have ever abandoned earplugs because they felt dry and intrusive rather than because they didn't block enough, this is the specific thing custom moulds fix.
- **Ask for material by use case, not one pair for everything.** Soft silicone for sleeping and lying on your side; hard acrylic for daytime. Buying one pair to cover both is the mistake this avoided.
- **Buy spares if you can afford to.** Ideally I would have a spare of each material. In practice I have one spare acrylic pair. These are small objects that live in a cabinet and get carried around; losing the only one at the wrong moment is the obvious failure mode.
- **Cost was ~USD 100 for the set of two**, and will vary a lot by country and provider. Treat it as an order of magnitude. Worth knowing that the price is what deferred this for years, and in hindsight the deferral was the expensive part.
- **Judge the provider by whether they will talk you out of a purchase.** The most useful thing mine did was tell me I did not need to buy anything else. A provider who sells you a third pair for every new situation is telling you something.

Earmuffs, worn over earplugs

kind: product · verdict: partial · since: unknown · cost: unknown · modality: auditory

Earmuffs, worn over earplugs

Personal experience, not evidence. Not medical advice — see the disclaimer in the *README*.

The problem

The top of the escalation ladder: situations where a closed door is not enough and earplugs alone are not enough either. In practice this has been someone else's television while I am trying to study or work.

What was tried

Earmuffs worn **on top of** earplugs, rather than instead of them — the doubling-up approach.

What happened

Tried sometimes, never adopted. **I have not found them as comfortable**, and earplugs remain my general go-to.

What it did not fix

- **Comfort.** Same failure mode that decided every other entry in this directory: the question is not what a thing blocks, it is whether you will keep it on.
- **No attenuation claim is being made either way.** Doubling up plausibly blocks more than plugs alone; I have not measured it and neither has this notebook. The verdict here is about wearability only.
- **Nothing is recorded about the specific muffs.** A different pair might sit differently. This is one person's experience of an unrecorded model, not a verdict on the category.

Transferability

- **Worth trying if plugs alone are not covering it**, since muffs-over-plugs is the standard way to stack protection and it costs nothing to test if you already own plugs.
- **Expect the comfort tax to be the deciding factor**, as it was here. Clamping force and heat are the usual complaints, and they compound with an earplug already in place.
- **If you get on with them, you have an option this notebook does not.** The absence of a positive verdict here is one person's ears, not a finding.

Loop earplugs

kind: product · verdict: partial · since: 2023-00 · cost: unknown · modality: auditory

Loop earplugs

Personal experience, not evidence. Not medical advice — see the disclaimer in the *README*.

The problem

Same problem as the custom moulds: honking at night in the previous Jerusalem apartment, and a search for anything that would help. Full background in *custom-moulded-earplugs.md*.

What was tried

Loop earplugs, bought before the custom moulded pair.

There was a second reason I came to them, separate from the noise. **Loop have done genuinely important work normalising sensory difficulties.** When they became popular, I started hearing from friends — people I would never have associated with sensory issues at all — that Loops had been life-changing for them. That surfacing effect is real and I have a lot of respect for it. The product got people talking about something they had been handling silently.

What happened

They work, and I like them. This is not a negative review.

But the custom moulds were significantly more comfortable, and comfort decides which pair you actually wear. The customs are the ones I have on all the time; the Loops are not. Once I had both, the choice made itself.

What it did not fix

- **Comfort, relative to a custom mould.** This is the entire gap. An earplug that is merely tolerable loses to one that is inert, every time, because the decision gets made fresh every day.
- **Nothing about attenuation is being claimed here.** I have not compared what the two actually block. The comparison I am making is about wearability, and it should not be read as a performance comparison.

Transferability

- **Start here rather than at the customs, if you have not tried anything.** They are far cheaper, they need no appointment, and for a lot of people — including friends of mine — they are the end of the search rather than a step in it.
- **The reason to escalate to custom moulds is comfort, not performance.** If Loops sit fine in your ears and you wear them, you have already got the result and can stop.
- **If you abandon them, notice why.** “Didn’t block enough” and “I couldn’t stand having them in” point at different next purchases.

Part IV — Sources

One entry per load-bearing source. The **read** field records how much of each was actually seen, so a verified claim can be told from a relayed one without redoing the work.

Auditory Processing Disorder: contested construct, heavy ADHD overlap

Auditory Processing Disorder in childhood: a critical appraisal of diagnostic validity, functional assessment, and interdisciplinary practice. *Frontiers in Human Neuroscience*. 2026;20:1715787.

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Atypical sensory profiles as core features of adult ADHD

Bijlenga D, Tjon-Ka-Jie JYM, Schuijers F, Kooij JJS. Atypical sensory profiles as core features of adult ADHD, irrespective of autistic symptoms. *European Psychiatry*. 2017 Jun;43:51–57.

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How loud a crying child actually is at a carer’s ear

Calderon LE, Carney LD, Kavanagh KT. The Cry of the Child and its Relationship to Hearing Loss in Parental Guardians and Health Care Providers. *Journal of Evidence-Informed Social Work*. 2016;13(2):198–205.

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What custom moulded plugs actually attenuate, and what sets the curve

Synthesis of manufacturer and clinical technical material on custom-moulded hearing protection, incl. Minuendo, ‘Custom Molded Earplugs — The Myths and Truths’; Honeywell Safety, ‘So, What’s the Story with Custom-Molded Earplugs?’; MEE Audio custom earplug filter specifications.

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Decreased sound tolerance: hyperacusis, misophonia, phonophobia — and what none of them are

Multiple — see body. Anchor: Aazh H, et al. Sound Tolerance Conditions (Hyperacusis, Misophonia, Noise Sensitivity, and Phonophobia): Definitions and Clinical Management. PMID 35858241.

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Perceptual load counters increased distraction in ADHD

Forster S, Robertson DJ, Jennings A, Asherson P, Lavie N. Plugging the attention deficit: Perceptual load counters increased distraction in ADHD. *Neuropsychology*. 2014;28(1):91–97.

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Speech intelligibility (STI), not loudness, predicts performance loss

Haapakangas A, Hongisto V, Liebl A. The relation between the intelligibility of irrelevant speech and cognitive performance — A revised model based on laboratory studies. *Indoor Air*. 2020;30(6):1130–1146.

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Does wearing earplugs help you concentrate? The evidence is thinner than the marketing

Synthesis: Smalt CJ et al., The Effect of Hearing-Protection Devices on Auditory Situational Awareness and Listening Effort (PMID 31045653); simulated temporary hearing deficit pilot study (PMC12922070); Effect of Hearing Protection Use on Pianists' Performance and Experience (PMC9315358).

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Overprotection: the reason “block as much as possible” is wrong

Synthesis of NIOSH Criteria for a Recommended Standard: Occupational Noise Exposure (DHHS/NIOSH Publication 98-126, 1998); CDC/NIOSH hearing protection guidance; and EPA, Product Noise Labeling — Hearing Protection Devices, proposed rule, 74 FR 39150, 2009-08-05.

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Hearing protection and whether you can still hear the alarm

Institut National de Recherche et de Sécurité (INRS) and CNRS UMR5292. Experimental assessment of the effect of wearing hearing protectors on the audibility of railway warning signals for normal hearing and hearing impaired listeners. International Journal of Occupational Safety and Ergonomics. 2022;28(4):2385–2395.

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STI 0.20 and STI 0.50: what the two standard thresholds actually mean

ISO 3382-3, Acoustics — Measurement of room acoustic parameters — Part 3: Open plan offices; with supporting irrelevant-speech experimental literature incl. Haapakangas et al. and Ebissou/Jeon-type STI-graded studies.

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The standard that already answers “can I hear an alarm through hearing protection”

ISO 7731:2003. Ergonomics — Danger signals for public and work areas — Auditory danger signals. 2nd edition, 2003-11-01. ISO/TC 159 Ergonomics, SC 5 Ergonomics of the physical environment.

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The Israeli siren level specification is not publicly documented

Negative result. Searches for the Israeli Home Front Command (פיקוד העורף) public siren sound-pressure-level specification, English and Hebrew, 2026-08-12.

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Perceptual abnormalities related to sensory gating deficit in adult ADHD

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Waking up is a different threshold from hearing

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How concentration shields against distraction

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Sensory processing disorder and sensory over-responsivity: what has standing and what doesn't

Multiple — see body. Anchors: American Academy of Pediatrics policy statement (2012); The Debate Over Sensory Processing Disorder, *American Journal of Psychiatry Residents' Journal* (2022), doi:10.1176/appi.ajp-rj.2022.170405.

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